

MATH70102 UNSUPERVISED LEARNING

LIVE SESSION — WEEK 3

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IMPERIAL

Review of this week's content

Computational examples

Problem sheet questions

Canonical correlation analysis

Overview

- For two multivariate data sets with matching observations, find **projections** that **maximise correlation**
- The projections are required to be **orthogonal** with respect to the other optimal projections of the same sample
- Find patterns that are **common** among the two data sets

Canonical correlation analysis (cont'd)

Data and basic sample statistics

- **Samples:** $\mathbf{x}_1, \dots, \mathbf{x}_n \in \mathbb{R}^d$ and $\mathbf{y}_1, \dots, \mathbf{y}_n \in \mathbb{R}^e$ (note: $d = e$ not required)
- **Sample means:** $\bar{\mathbf{x}} \in \mathbb{R}^d$ and $\bar{\mathbf{y}} \in \mathbb{R}^e$
- **Sample covariance matrices:** $\Sigma_{\mathbf{x}} \in \mathbb{R}^{d \times d}$ and $\Sigma_{\mathbf{y}} \in \mathbb{R}^{e \times e}$
- **Sample cross-covariance matrix:**

$$\Sigma_{\mathbf{xy}} := \frac{1}{n} \sum_{i=1}^n (\mathbf{x}_i - \bar{\mathbf{x}})(\mathbf{y}_i - \bar{\mathbf{y}})^T \in \mathbb{R}^{d \times e}$$

(contains all cross-covariances between the components of two data sets)

Canonical correlation analysis (cont'd)

Projections and correlation

- **Projections** of the two data sets via $\mathbf{a} \in \mathbb{R}^d$ and $\mathbf{b} \in \mathbb{R}^e$:

$$\mathbf{a}^T \mathbf{x}_i, \quad i = 1, \dots, n, \quad \mathbf{b}^T \mathbf{y}_i, \quad i = 1, \dots, n$$

- **Sample correlation** between the projections:

$$\begin{aligned} \rho &:= \rho(\mathbf{a}, \mathbf{b}) := \frac{\frac{1}{n} \sum_{i=1}^n (\mathbf{a}^T \mathbf{x}_i - \mathbf{a}^T \bar{\mathbf{x}}) (\mathbf{b}^T \mathbf{y}_i - \mathbf{b}^T \bar{\mathbf{y}})}{\sqrt{\frac{1}{n} \sum_{i=1}^n (\mathbf{a}^T \mathbf{x}_i - \mathbf{a}^T \bar{\mathbf{x}})^2} \sqrt{\frac{1}{n} \sum_{i=1}^n (\mathbf{b}^T \mathbf{y}_i - \mathbf{b}^T \bar{\mathbf{y}})^2}} \\ &= \frac{\mathbf{a}^T \Sigma_{xy} \mathbf{b}}{\sqrt{\mathbf{a}^T \Sigma_x \mathbf{a}} \sqrt{\mathbf{b}^T \Sigma_y \mathbf{b}}} \in [-1, 1] \end{aligned}$$

Canonical correlation analysis (cont'd)

Canonical variables

- First pair of **canonical variables** is defined by

$$u_{1,i} := \mathbf{a}_1^T \mathbf{x}_i, \quad i = 1, \dots, n, \quad v_{1,i} := \mathbf{b}_1^T \mathbf{y}_i, \quad i = 1, \dots, n,$$

where $\mathbf{a}_1 \in \mathbb{R}^d$ and $\mathbf{b}_1 \in \mathbb{R}^e$ maximise $\rho(\mathbf{a}_1, \mathbf{b}_1)$

- Unique only up to **scaling by a positive factor** since

$$\rho(\alpha \mathbf{a}_1, \beta \mathbf{b}_1) = \rho(\mathbf{a}_1, \mathbf{b}_1), \quad \alpha, \beta > 0$$

Canonical correlation analysis (cont'd)

Canonical variables

- k -th pair of **canonical variables** (for $k \geq 2$) is defined by

$$u_{k,i} := \mathbf{a}_k^T \mathbf{x}_i, \quad i = 1, \dots, n, \quad v_{1,i} := \mathbf{b}_k^T \mathbf{y}_i, \quad i = 1, \dots, n,$$

where $\mathbf{a}_k \in \mathbb{R}^d$ and $\mathbf{b}_k \in \mathbb{R}^e$ maximise $\rho(\mathbf{a}_k, \mathbf{b}_k)$ such that

$$\mathbf{a}_k^T \Sigma_{\mathbf{x}} \mathbf{a}_j = 0 \quad \text{for any } j < k,$$

$$\mathbf{b}_k^T \Sigma_{\mathbf{y}} \mathbf{b}_j = 0 \quad \text{for any } j < k$$

orthogonality

- We can find up to $\min\{d, e\}$ pairs ($k \leq \min\{d, e\}$)

Canonical correlation analysis (cont'd)

Solution

- **Assumption:** $\Sigma_{\mathbf{x}}$ and $\Sigma_{\mathbf{y}}$ **invertible**
- Let $k = 1, \dots, \min\{d, e\}$
- $\mathbf{a}_k$ is the **eigenvector** of

$$\Sigma_{\mathbf{x}}^{-1} \Sigma_{\mathbf{xy}} \Sigma_{\mathbf{y}}^{-1} \Sigma_{\mathbf{xy}}^T$$

corresponding to its **k -th largest eigenvalue**

- $\mathbf{b}_k := \Sigma_{\mathbf{y}}^{-1} \Sigma_{\mathbf{xy}}^T \mathbf{a}_k$

Non-negative matrix factorisation

Overview

- For a matrix $X \in \mathbb{R}_{\geq 0}^{d \times n}$ and $q \leq \max\{d, n\}$, try to find $W \in \mathbb{R}_{\geq 0}^{d \times q}$ and $H \in \mathbb{R}_{\geq 0}^{q \times n}$ such that

$$X \approx WH$$

- Write $X = [\mathbf{x}_1 \ \cdots \ \mathbf{x}_n]$ for $\mathbf{x}_1, \dots, \mathbf{x}_n \in \mathbb{R}^d$ and $H = [\mathbf{h}_1 \ \cdots \ \mathbf{h}_n]$ for $\mathbf{h}_1, \dots, \mathbf{h}_n \in \mathbb{R}^q$
- Then $W\mathbf{h}_i$ is a **reconstruction** of $\mathbf{x}_i$ from its **lower-dimensional description** $\mathbf{h}_i$
- Monotonicity:**

increase in $H \quad \Rightarrow \quad$ increase in X

Non-negative matrix factorisation (cont'd)

Quantifying $X \approx WH$

- Euclidean distance (to **minimise**):

$$\sum_{i=1}^d \sum_{j=1}^n (X_{i,j} - (WH)_{i,j})^2$$

- Quasi-likelihood function (to **maximise**):

$$L(W, H) := \sum_{i=1}^d \sum_{j=1}^n \left(X_{i,j} \log((WH)_{i,j}) - (WH)_{i,j} \right),$$

(summand is log likelihood of $X_{i,j} \sim \text{Poisson}((WH)_{i,j})$)

Non-negative matrix factorisation (cont'd)

Algorithm

To maximise $L(W, H)$, **iterate** updates:

$$W_{i,k} \leftarrow W_{i,k} \frac{\sum_{j=1}^n H_{k,j} X_{i,j} / (WH)_{i,j}}{\sum_{j=1}^n H_{k,j}}, \quad i = 1, \dots, d, \quad k = 1, \dots, q$$

$$H_{k,j} \leftarrow H_{k,j} \frac{\sum_{i=1}^d W_{i,k} X_{i,j} / (WH)_{i,j}}{\sum_{i=1}^d W_{i,k}}, \quad k = 1, \dots, q, \quad j = 1, \dots, n$$

Initial values for W and H : data-driven or randomised

Non-negative matrix factorisation (cont'd)

Caveats

- Algorithm may converge to **local optimum** (try different initial values)
- Factorization is **not unique** (even in the ideal case $X = WH$)

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Problem 1

Question

The data set `simdata.csv` contains 1500 samples in $\mathbb{R}^{100}$ generated by simulation. (Each row is one sample.) Is it possible to distinguish **group structure** within the samples by:

- (a) drawing two-dimensional scatterplots of some pairs of components?
 - (b) applying ordinary PCA?
 - (c) applying kernel PCA?
- (In (b) and (c), it is useful to draw scatterplots of the highest order principal components.)

Problem 1

Solution

→ Jupyter Notebook

Problem 2

Question

Show that the projection vectors $\mathbf{a}_k$ and $\mathbf{a}_j$ in the solution to canonical correlation analysis (p. 20 in the Lecture Notes) indeed satisfy the constraint

$$\mathbf{a}_k^T \Sigma_{\mathbf{x}} \mathbf{a}_j = 0$$

when $j < k$. *Hint:* Consider $\tilde{\mathbf{a}}_k := \Sigma_{\mathbf{x}}^{1/2} \mathbf{a}_k$ and $\tilde{\mathbf{a}}_j := \Sigma_{\mathbf{x}}^{1/2} \mathbf{a}_j$.

Problem 2

Solution

According to the CCA solution, $\tilde{\mathbf{a}}_k = \Sigma_{\mathbf{x}}^{1/2} \mathbf{a}_k$ and $\tilde{\mathbf{a}}_j = \Sigma_{\mathbf{x}}^{1/2} \mathbf{a}_j$ are **eigenvectors** of the matrix

$$\Sigma_{\mathbf{x}}^{-1/2} \Sigma_{\mathbf{xy}} \Sigma_{\mathbf{y}}^{-1} \Sigma_{\mathbf{xy}}^T \Sigma_{\mathbf{x}}^{-1/2}$$

Thus, they are **orthogonal**, whereby

$$\begin{aligned} 0 &= \tilde{\mathbf{a}}_k^T \tilde{\mathbf{a}}_j = (\Sigma_{\mathbf{x}}^{1/2} \mathbf{a}_k)^T (\Sigma_{\mathbf{x}}^{1/2} \mathbf{a}_j) \\ &= \mathbf{a}_k^T \Sigma_{\mathbf{x}}^{1/2} \Sigma_{\mathbf{x}}^{1/2} \mathbf{a}_j \\ &= \mathbf{a}_k^T \Sigma_{\mathbf{x}} \mathbf{a}_j \end{aligned}$$

Problem 3

Question

- (a) (Non-existence of exact NMF) Show that matrices $W \in \mathbb{R}_{\geq 0}^{2 \times 1}$ and $H \in \mathbb{R}_{\geq 0}^{1 \times 2}$ such that

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = WH$$

do not exist. *Hint:* Write W and H element by element and study the equations they would have to satisfy.

Problem 3

Solution

Let

$$W := \begin{bmatrix} w_1 \\ w_2 \end{bmatrix}, \quad H := \begin{bmatrix} h_1 & h_2 \end{bmatrix}.$$

Then, $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = WH$ is equivalent to

$$w_1 h_1 = 1, \quad w_1 h_2 = 0, \quad w_2 h_1 = 0, \quad w_2 h_2 = 1.$$

But the two equalities in the middle imply ($w_1 = 0$ or $h_2 = 0$)
and ($w_2 = 0$ or $h_1 = 0$)

Which in turn implies $w_1 h_1 = 0$ or $w_2 h_2 = 0 \Rightarrow$ **contradiction**

Problem 3

Question

(b) (Non-uniqueness of NMF) Show that there are multiple $W \in \mathbb{R}_{\geq 0}^{2 \times 1}$ and $H \in \mathbb{R}_{\geq 0}^{1 \times 2}$ such that

$$\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} = WH.$$

Problem 3

Solution

$$W := \begin{bmatrix} \alpha \\ \alpha \end{bmatrix}, \quad H := \begin{bmatrix} \alpha^{-1} & \alpha^{-1} \end{bmatrix}$$

for any $\alpha > 0$ will work